

Practical AI for Academics

Instructor: Claes Bäckman | **Institute:** Leibniz Institute SAFE

The course runs across three sessions spanning Thursday afternoon and Friday. Thursday goes through the conceptual foundation to explain what these tools are and how they work. Friday morning goes through concrete workflows with examples related to pre-submission referee reports on your own paper, making slides, and other examples. A third block on Friday afternoon is held open for further hands-on work if participants want it.

Day 1 • Thursday

15:00 -- 17:00+

Concepts and Tools

- 01 **The landscape** — What an LLM is and an overview of recent developments.
- 02 **Use cases** — Research, coding, teaching, admin
- 03 **How the tools work** — Chat vs. agentic tools, the harness, context windows, pricing and model tiers.
- 04 **In practice** — Workflows, the importance of context, fetching public data via APIs.
- 05 **Verification & what's next** — How to verify output and getting set up for Friday.

Day 2, Block 1 • Friday morning

09:00/10:00 -- 12:00

Practical Workflows

- 01 **Workspace setup** — Claude Code in the app, VS Code + Claude Code, CLAUDE.md, Overleaf
- 02 **Referee report** — Live demo of a skill to give you feedback on your paper
- 03 **Voice file** — Extract your own prose style from past writing so edits sound like you.
- 04 **Paper & code** — Checking whether the code actually does what the paper claims.
- 05 **Slides** — Building Beamer decks from a reusable style file.
- 07 **Teaching** — Protecting learning from AI, then using AI to teach more.

Day 2, Block 2 • Friday afternoon

14:00 -- 16:00

Hands-on Examples & Help (IF NEEDED)

For participants who want more time applying the morning's workflow to their own papers, or who hit something specific worth troubleshooting together.